Lab # 02

COURSE CODE: SE-206L

COURSE TITLE:

Software Construction & Development

LAB TITLE

Semester Project Selection & UML Architecture

LAB ASSESSMENT TABLE:

Obtained Marks	Total Marks

Submitted To: Engr. Rizwan Shah.

Submitted By: Muhammad Danial.

Registration – No: 24ABSWE0021.

Semester: 5th

Date: 29-09-2026.

Dept. of Software Engineering

UET Peshawar, Abbottabad Campus.

Instructor Sign : _____

Lab 2:

Semester Project Selection & UML Architecture

Part 1: Semester Project Selection

Project Name: **Smart Inventory & Sales Management System.**

Introduction:

The **Smart Inventory & Sales Management System** is a software application designed to simplify business record management and provide a centralized platform for handling daily inventory and sales operations. It will be used by **Administrators, Managers, and Sales Staff**, with access based on their respective responsibilities.

Objectives

- Create one platform for handling inventory and sales-related activities.
- Maintain organized records of products, customers, and suppliers.
- Keep stock quantities updated automatically after purchases and sales.
- Control system access according to the responsibilities of each user.
- Reduce mistakes caused by manual record keeping.
- Notify users when products require restocking.
- Provide meaningful reports to support business monitoring and decision-making.
- Protect stored information through proper validation and error handling.

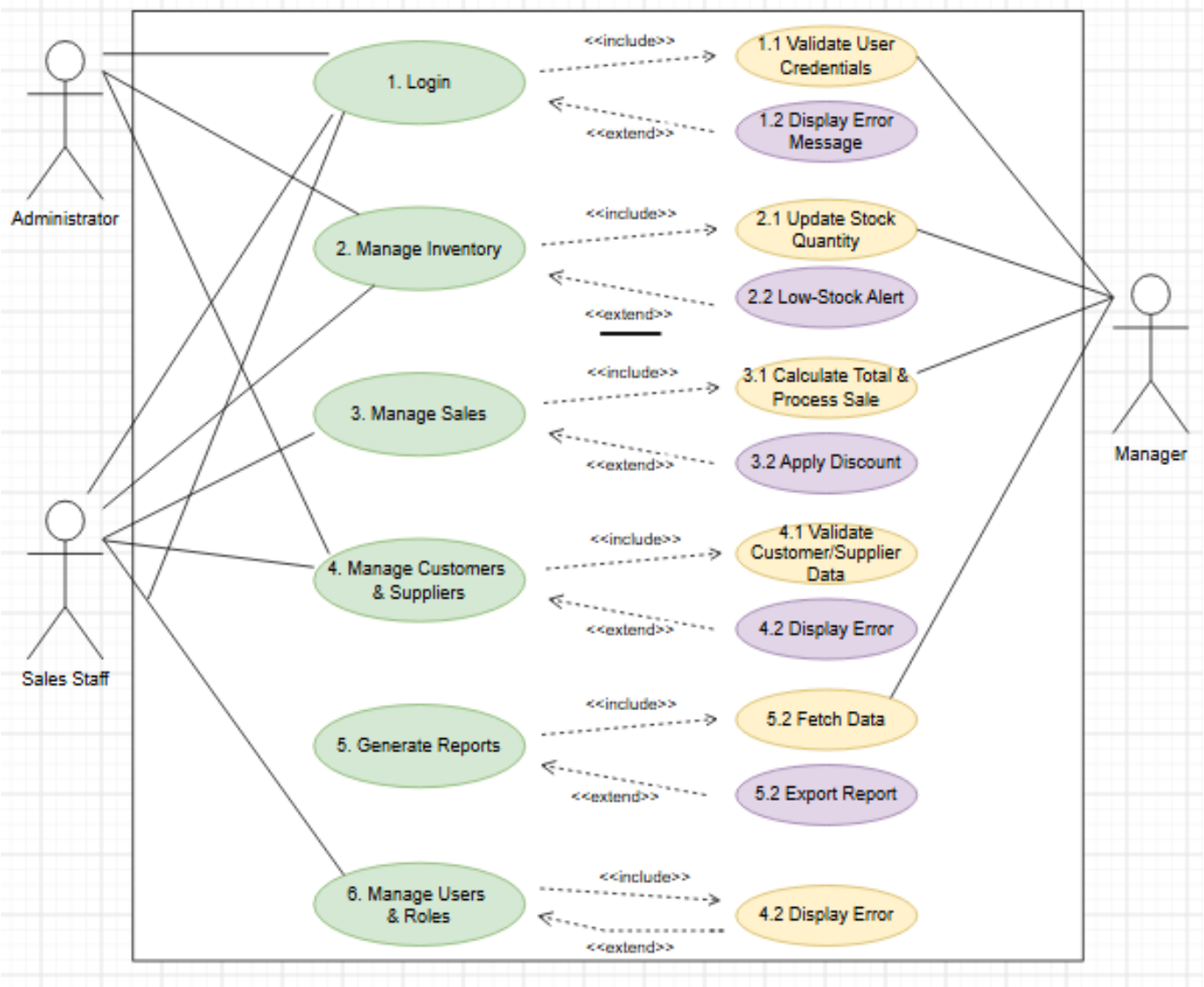
Key Metrics

- **Inventory Accuracy:** Measures the accuracy of stock quantities maintained by the system.
- **Sales Accuracy:** Measures the correctness of recorded sales transactions.
- **System Performance:** Measures how quickly the system responds to user actions.
- **Data Reliability:** Measures the consistency and correctness of stored business data.
- **Access Control:** Ensures users can access only the features assigned to their roles.
- **Low-Stock Alerts:** Measures how accurately the system identifies items needing restocking.
- **Report Accuracy:** Measures the correctness of information presented in generated reports.
- **Error Rate:** Tracks the frequency of invalid or unsuccessful system operations.

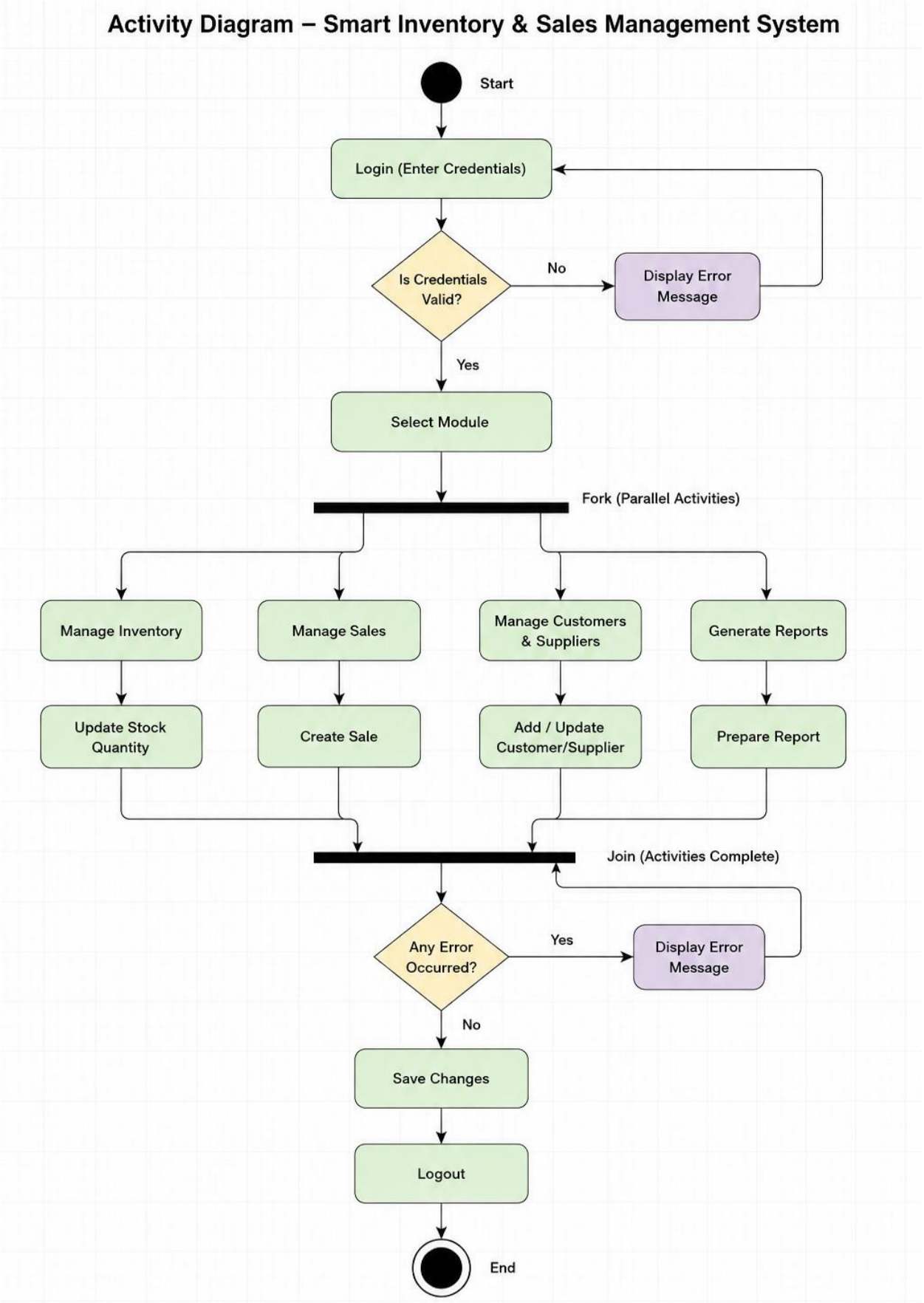
Part 2: Lab Tasks & UML Modeling

All diagrams are designed (I can share .drawio files if required)

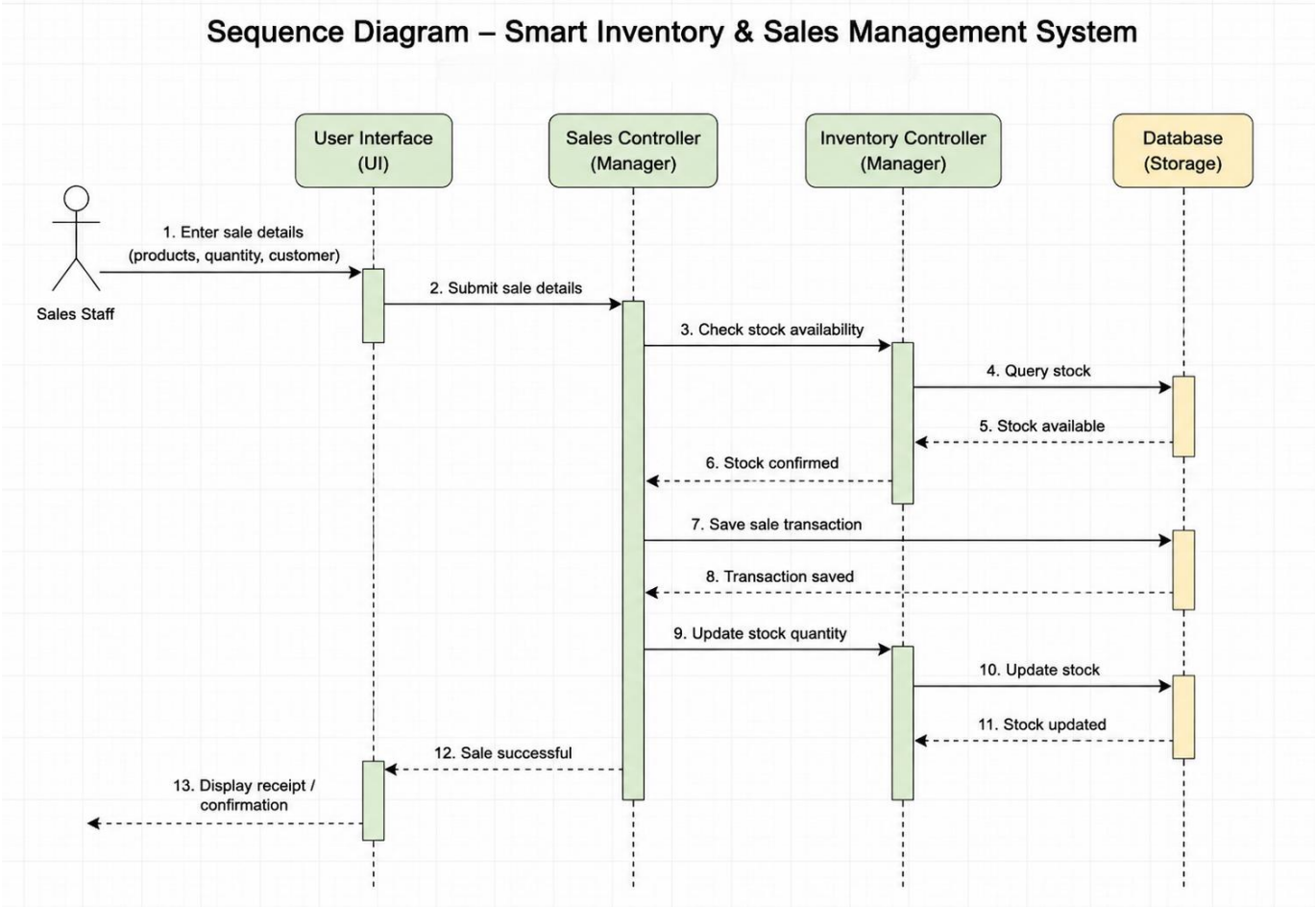
Task 1: Use Case Diagram (The High-Level View)



Task 2: Activity Diagram (The Logical Flow)



Task 3: Sequence Diagram (Chronological Interaction)



Task 4: Class Diagram (The Structural Blueprint)

